

T(2nd Sm.)-Computer Sc.-H/CC-3/CBCS

2021

COMPUTER SCIENCE — HONOURS

Paper : CC-3

(Data Structure)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

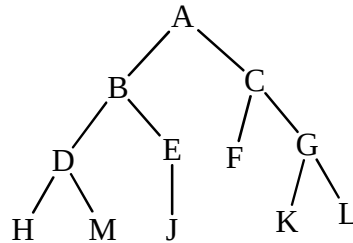
Answer *question No. 1* and *any four* from the rest.

1. Answer ***any five*** questions: 2×5
- (a) Give the formal definition of a linear array.
 - (b) Write the formula to find the location of any element of a one dimension linear array having base address (BA) and n number of words per memory cell.
 - (c) Represent the following polynomial using linked list:
$$P(x) = 2x^6 - 3x^5 + 7x - 8.$$

[Note: The list must have a header node.]
 - (d) Define a deque. What are its two variations?
 - (e) Convert the following postfix expression P into its equivalent infix for m . Then evaluate the infix expression. [Convert by inspection]
 $P : 16, 7, 3, -, /, 8, 4, 5, +, *, +.$
 - (f) State the two main properties of a well defined recursive procedure.
 - (g) Draw a complete binary tree having 28 nodes.
 - (h) Best case running time of Quick Sort is $O(n \log n)$ — Justify.
2. (a) Build a heap H from the following list of numbers:
54, 40, 60, 22, 70, 56, 87, 35.
- (b) Suppose LIST is a linked list in the memory. Write an algorithm to delete the last node from LIST. 5+5
3. (a) Write an algorithm to find an ITEM from a sorted list of elements.
- (b) Show the steps to sort the following list of numbers in ascending order using QUICKSORT:
46, 52, 35, 76, 40, 27, 16, 38, 63, 15. 5+5

Please Turn Over

4. (a) What are sparse matrices? Illustrate with an example a lower triangular matrix.
(b) Name the three standard algorithms used to traverse a binary tree. Traverse the following tree in the above mentioned methods:



3+(1+6)

5. (a) Why are threads used in binary trees?
(b) What are the properties of a BST? Write the steps to insert an ITEM in a BST at its appropriate place.
2+(2+6)
6. (a) State the names of different hashing functions used generally. Explain them briefly.
(b) How is collision resolved using open addressing?
(c) State the main disadvantage of linear probing.
6+2+2
7. (a) Write an algorithm to implement Merge Sort.
(b) Write an algorithm to insert nodes in a queue represented by a linked list.
6+4
8. (a) Illustrate with proper explanation, a technique that will minimize the overflow in a stack maintained in an array.
(b) What are priority queues? How are they maintained in memory?
6+(2+2)
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